

Kohn–Sham Theory from Effective Field Theory: Quasiparticle Band Narrowing from Frozen Core Dynamics

[Authors]¹

¹[Affiliation]

(Dated: April 17, 2026)

Kohn–Sham (KS) eigenvalues are routinely compared with angle-resolved photoemission (ARPES) and used as input for many-body methods, yet density functional theory (DFT) assigns them no physical meaning. For alkali and alkaline-earth metals, KS bandwidths overestimate ARPES measurements by 20–35%, a discrepancy that persists across all exchange-correlation functionals. Starting from the full electron–ion action, we construct an effective field theory (EFT) of the inhomogeneous electron gas that provides the missing justification: two conditions—a scale separation between core excitation energies and the valence Fermi energy, and an approximate Galilean invariance of the uniform electron gas confirmed by diagrammatic Monte Carlo—together imply that KS bands *are* the quasiparticle bands, up to a core-induced renormalization factor z^{core} . This factor, absent from standard DFT and conventional pseudopotential theory, reflects many-body dynamics of core electrons that conventional pseudopotentials freeze out and no static exchange-correlation potential can capture. It is of order unity for alkali metals and below 5% for Al and Si, directly explaining both the failure and success of KS band theory. We derive a closed-form post-SCF formula and apply it to Li, Na, K, Ca, Mg, Al, and Si. The predicted quasiparticle bands resolve the long-standing ARPES bandwidth discrepancy, matching the accuracy of embedded dynamical mean-field theory at negligible computational cost.

The Kohn–Sham (KS) Hamiltonian is the workhorse of modern electronic structure theory. For *sp*-bonded metals and semiconductors, its eigenvalues successfully approximate quasiparticle energies measured by angle-resolved photoemission spectroscopy (ARPES) [5, 6], and this agreement is widely taken for granted. Yet within density functional theory (DFT) the KS eigenvalues are purely auxiliary [7, 8]: there is no formal guarantee that they should approximate quasiparticle energies at all. For alkali and alkaline-earth metals this gap between practice and theory becomes visible: KS bandwidths systematically overestimate ARPES measurements by 20–35% [1–3], a discrepancy first noted in the 1980s that persists regardless of the exchange-correlation functional [9]. That only all-electron embedded dynamical mean-field theory (eDMFT) [9] resolves it—a many-body method that treats dynamical correlations beyond DFT—signals that the missing physics is not a better functional but something outside the KS framework altogether.

What is missing is a first-principles theory that establishes when and why KS eigenvalues approximate quasiparticle energies. DFT itself is not at fault: it provides a rigorous variational principle for the ground-state density [10, 11]. The gap lies in the step from DFT to the KS Hamiltonian, whose eigenvalues are formally auxiliary [7, 8] yet routinely used as physical, both for interpreting ARPES and as the starting point for *GW*, DMFT, and DFT+*U*.

Starting from the Born–Oppenheimer action for electrons in a lattice potential with Coulomb interactions, we construct an effective field theory (EFT) of the inhomogeneous electron gas, guided by two inputs: (i) a scale separation between core excitation energies and the valence

Fermi energy that allows core electrons to be integrated out, and (ii) locality properties of the interacting uniform electron gas (UEG) at metallic densities, established by high-order diagrammatic Monte Carlo (DiagMC) calculations, that dictate how many-body effects are absorbed into renormalized parameters. The tree-level quasiparticle propagator of this EFT turns out to be

$$G_{\nu\mathbf{k}}^{-1}(i\omega) \approx (z^{\text{val}})^{-1} [(z_{\nu\mathbf{k}}^{\text{core}})^{-1} i\omega - (\epsilon_{\nu\mathbf{k}}^{\text{KS}} - \epsilon_F)], \quad (1)$$

where, remarkably, the quasiparticle dispersion is governed by the eigenvalues $\epsilon_{\nu\mathbf{k}}^{\text{KS}}$ of the Kohn–Sham Hamiltonian $\hat{H}_{\text{KS}} = -\nabla^2/2 + V_{\text{PSP}} + V_H[n] + V_{xc}[n]$. Here z^{val} is the valence quasiparticle residue and $z_{\nu\mathbf{k}}^{\text{core}}$ is a closed-form frozen-core renormalization determined by the core form factors f_K^c and excitation energies ΔE_c , given explicitly by Eq. (3) below. In Eq. (1), z^{val} is an overall prefactor that does not shift the pole, while $z_{\nu\mathbf{k}}^{\text{core}}$ rescales the frequency relative to the dispersion—an effective time dilation caused by virtual core excitations. Locating the pole gives the quasiparticle energy measured by ARPES,

$$\epsilon_{\nu\mathbf{k}}^{\text{QP}} \approx \epsilon_F + z_{\nu\mathbf{k}}^{\text{core}} (\epsilon_{\nu\mathbf{k}}^{\text{KS}} - \epsilon_F), \quad (2)$$

so that the Kohn–Sham eigenvalues *are* the quasiparticle energies, compressed toward ϵ_F by the state-dependent factor $z_{\nu\mathbf{k}}^{\text{core}}$. Equation (1) is not an ansatz: it is the unique tree-level consequence of conditions (i) and (ii). The frozen-core factor z^{core} , absent from DFT and overlooked by conventional pseudopotential theory [12, 13], is of order unity in alkali and alkaline-earth metals (where ΔE_c is only a few Hartree) and < 5% in *sp*-bonded metals and semiconductors, and is therefore the direct source of the 20–35% alkali-metal bandwidth discrepancy introduced above.

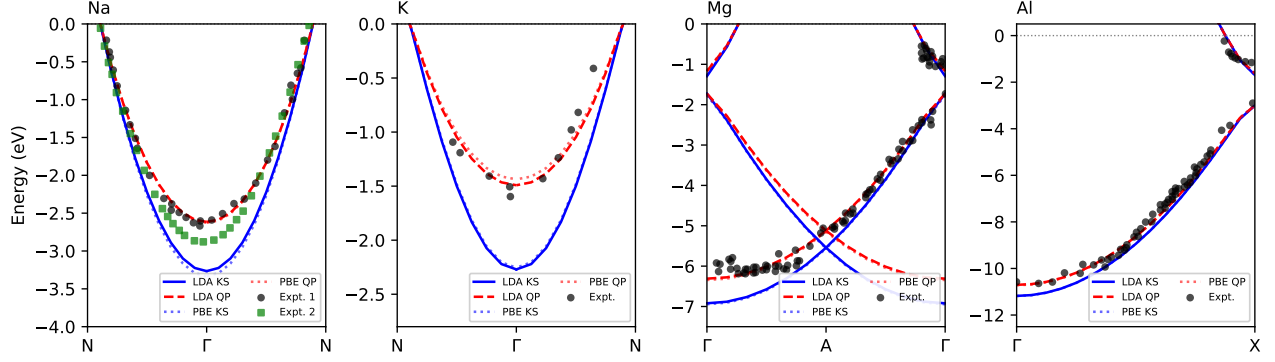


FIG. 1. Valence bands compared with ARPES data. Solid: LDA KS (blue) and QP (red). Dotted: PBE KS and QP. Na and K along N- Γ -N: Expt. 1 = Ref. [1], Expt. 2 = Ref. [2], K expt. = Ref. [3]. Mg along Γ -A- Γ : expt. = Ref. [4]. Al along Γ -X: expt. = Ref. [5]. PBE closely tracks LDA, confirming robustness to the functional.

Prerequisite (i) is the scale separation between the core excitation energies $\Delta E_c \sim 2\text{--}100$ Ha and the valence Fermi energy $\epsilon_F \sim 0.1$ Ha—three orders of magnitude apart. It is handled by a dual-fermion transformation [14, 15], which integrates out the core in a controlled expansion with small parameter $\epsilon_F/\Delta E_c \lesssim 0.05$. The result is a valence-only propagator whose static part reproduces a conventional pseudopotential [16–18] and whose dynamical part is separable in Bloch momenta. Solving the Dyson equation for this separable self-energy gives the closed-form frozen-core renormalization appearing in Eqs. (1)–(2),

$$z_{\nu\mathbf{k}}^{\text{core}} = \frac{1}{1 + \sum_c |F_{\nu\mathbf{k}}^c|^2 / \Delta E_c^2}, \quad F_{\nu\mathbf{k}}^c = \sum_{\mathbf{G}} c_{\nu\mathbf{k}}(\mathbf{G}) f_{|\mathbf{k}+\mathbf{G}|}^c, \quad (3)$$

where f_K^c is the core form factor [Eq. (7)], ΔE_c the core excitation energy of channel c , and $c_{\nu\mathbf{k}}(\mathbf{G})$ the Bloch coefficients of the valence eigenstates.

Prerequisite (ii), the approximate Galilean invariance, is more subtle. DiagMC data for the UEG at metallic densities [19] (Fig. 3) show that the interacting self-energy depends on $(i\omega, \mathbf{k})$ approximately only through the Galilean combination $i\omega - k^2/(2m)$: the quasiparticle residue $z(k)$ is nearly constant and $m^* \approx m$ throughout the occupied Fermi ball. This means that the insights of the Polchinski-Shankar Fermi-surface RG [20, 21]—in particular the hierarchy of relevant, marginal, and irrelevant couplings—extend approximately to the entire Fermi sea, directly prescribing how many-body effects should be absorbed into renormalized parameters. The resulting renormalized perturbation theory (RPT) for the inhomogeneous valence liquid yields the tree-level quasiparticle propagator Eq. (1) and thereby explains the origin and physical meaning of the KS Hamiltonian. The $(z^{\text{core}})^{-1}$ factor survives because core electrons do not screen dynamically ($\Gamma_3^{\text{core}} \approx 1$, polarizability suppressed by $\epsilon_F/\Delta E_c \sim 0.02\text{--}0.05$). We validate Eq. (2) against

eDMFT and ARPES for Li, Na, K, Ca, Mg, Al, and Si (Table I, Fig. 1).

Integrating out the core.—The first EFT step starts from the electron-ion Lagrangian ($S = \int_0^\beta d\tau L$),

$$L = \bar{\psi} g_0^{-1} \psi + L_{\text{int}}, \quad (4)$$

where $g_0^{-1} = \partial_\tau - \nabla^2/2 + V_{\text{lat}} - \mu$ is the bare single-particle propagator, $L_{\text{int}} = \frac{1}{2} \int_{\mathbf{r}\mathbf{r}'} \bar{\psi}_{\mathbf{r}} \bar{\psi}_{\mathbf{r}'} v(\mathbf{r}-\mathbf{r}') \psi_{\mathbf{r}'} \psi_{\mathbf{r}}$ is the Coulomb interaction, and $\psi A \equiv \int_{\mathbf{r}\mathbf{r}'} \bar{\psi}_{\mathbf{r}} A(\mathbf{r}, \mathbf{r}') \psi_{\mathbf{r}'}$. The hierarchy $\epsilon_F \ll \Delta E_c$ gives a controlled expansion with small parameter $\epsilon_F/\Delta E_c \lesssim 0.05$: integrating out the core Grassmann fields via a dual-fermion transformation [14, 15] (Appendices A–B) produces a valence effective action S_{val} ,

$$L_{\text{val}} = \bar{\psi} (g^{\text{val}})^{-1} \psi + L_{\text{int}}^{\text{val}}, \quad (5)$$

with bare valence propagator $(g^{\text{val}})^{-1} = (z^{\text{core}})^{-1} \partial_\tau - \nabla^2/2 + V_{\text{PSP}} - \mu$ (where V_{PSP} is the nonlocal static pseudopotential [16–18, 22]), $L_{\text{int}}^{\text{val}}$ the Coulomb interaction among valence electrons, and a dynamical part. The full pseudopotential splits as

$$(\delta V_{pp})_{st} = \Sigma_{st}^{\text{HF}} + \sum_c \frac{M_{sc} M_{tc}^*}{i\omega + \Delta E_c} + O(\epsilon_F^2/\Delta E_c), \quad (6)$$

where s, t label valence orbitals, Σ^{HF} is the static Hartree-Fock self-energy of the core (reproducing conventional norm-conserving PSPs [16–18, 22]; the formal vertex residual from the $z\Gamma_3$ cancellation for nonlocal Kleinman-Bylander projectors is absorbed by the fitted PSP parameters), and the dynamical piece has poles at the core excitation energies ΔE_c with matrix elements M_{sc} coupling valence orbital s to core hole c (Appendix B).

In momentum space the dynamical term is separable, $V_{\text{dyn}}(\mathbf{K}, \mathbf{K}'; i\omega) = \sum_c f_K^c f_{K'}^c / (i\omega + \Delta E_c)$, with the Wilson coefficient

$$f_K^c = \frac{\sqrt{4\pi}}{K} \int_0^\infty u_c(r) [V_{H,c}(r) - J_c] \sin(Kr) dr \quad (7)$$

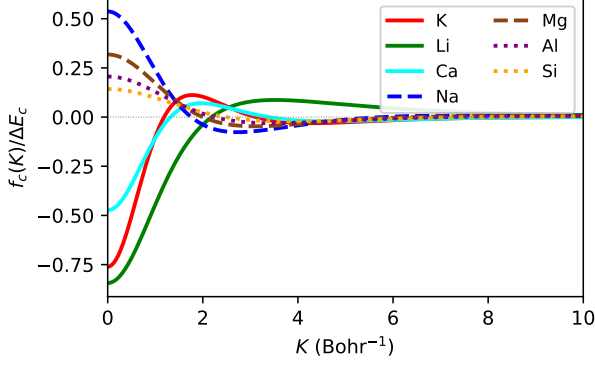


FIG. 2. Dimensionless ratio $f_K^c / \Delta E_c$ for the dominant core channel of each element. This ratio controls z^{core} [Eq. (3)]: elements with large $|f_K^c / \Delta E_c|$ have large corrections. K and Li have the largest ratios because ΔE_c is small (1.7 to 2.6 Ha) while $|f_K^c| \sim J_c \sim 1$ to 2 Ha is comparable. The [Ne]-core series Na–Mg–Al–Si shows monotonically decreasing amplitude: $|f_K^c|$ is nearly the same for all four because J_c and the orbital size $\langle r \rangle_c$ scale inversely with Z , keeping $|f_c| \sim J_c \langle r \rangle_c \sim 1$ Ha; meanwhile ΔE_{2s} roughly triples from Na to Si. The sign change near $K \approx 2$ to 3 Bohr^{-1} leads to partial cancellation in the coherent sum $F_{\nu\mathbf{k}}^c$.

computed from atomic Kohn–Sham core wavefunctions (Appendix H). Here $u_c(r) = rR_c(r)$ is the reduced radial core wavefunction, $V_{H,c}$ its orbital Hartree potential, and J_c the self-Coulomb integral. Solving the Dyson equation for this separable self-energy gives the closed-form $z_{\nu\mathbf{k}}^{\text{core}}$ of Eq. (3) (Appendix D). Figure 2 shows the dimensionless ratio $f_K^c / \Delta E_c$ that controls the magnitude of the frozen-core correction via $1 - z_{\Gamma}^{\text{core}} \approx (f_0^c / \Delta E_c)^2$ near Γ .

From the valence EFT to Kohn–Sham.—The second step constructs a renormalized perturbation theory (RPT) for the valence EFT, Eq. (5). The Polchinski–Shankar RG [20, 21] for a Fermi liquid classifies couplings into three categories: the local chemical potential is *relevant* (must be matched exactly), the quasiparticle residue z and effective mass m^* are *marginal* (should be matched as accurately as possible), and all remaining couplings—gradients, lattice nonlocality, residual vertex structure—are *irrelevant* at leading order. This hierarchy is formulated near the Fermi surface. The DiagMC data discussed above (Fig. 3) show that it extends to the entire occupied Fermi ball: $z(k)$ is nearly constant and $m^* \approx m$, so a single matching at the Fermi surface fixes the renormalization of all occupied states. The Ward identity $z\Gamma_3 = 1$ further ties the charge-vertex dressing to the residue, so matching z simultaneously matches the charge renormalization from electron–hole screening. DiagMC data [19] confirm that this cancellation also persists throughout the Fermi ball. For the inhomogeneous valence liquid, the locality of these properties means that the Fermi-surface matching can be performed locally, at

each point in real space: this is what enables a local density approximation (LDA), matching the RPT parameters to a local Fermi liquid at density $n(\mathbf{r})$.

Relevant: density matching.—The tree-level quasiparticle propagator must reproduce $n(\mathbf{r})$. This fixes the static self-energy at the Fermi level, yielding the LDA exchange-correlation potential as a matching condition:

$$V_{xc}^{\text{LDA}}[n](\mathbf{r}) \equiv \Sigma_{xc}^0(n(\mathbf{r})), \quad \Sigma_{xc}^0(n) = \Sigma_{xc}^{\text{UEG}}(n; k_F, i0^+). \quad (8)$$

The LDA functional is thus not an approximation to an unknown functional but the exact relevant matching condition of the EFT.

Marginal: residue matching.—The quasiparticle residue z_{Γ}^{val} is fixed by the slope of the UEG self-energy at the Fermi level. Since integrating out the core has already rescaled the frequency to $\omega' = (z^{\text{core}})^{-1}\omega$, and the approximate Galilean invariance of the UEG self-energy ensures that it depends on $i\omega'$ and $\mathbf{k}$ only through $i\omega' - \xi_{\nu\mathbf{k}}$ ($\xi_{\nu\mathbf{k}} = |\mathbf{k} + \mathbf{G}|^2/2m - \epsilon_F$), the projection of the valence self-energy onto the matched subspace is

$$P_2^{\Gamma} \Sigma_{\nu\mathbf{k}}^{\text{val}}(i\omega') \equiv V_H[n] + \Sigma_{xc}^0(n) + [1 - (z_{\Gamma}^{\text{val}})^{-1}](i\omega' - \xi_{\nu\mathbf{k}}), \quad (9)$$

where P_2^{Γ} extracts the relevant and marginal components (Appendix E).

Tree-level Lagrangian.—The matching conditions [Eqs. (8) and (9)] complete the RPT: the matched relevant and marginal couplings are absorbed into the propagator as counterterms, leaving the residual $\delta\Sigma = (1 - P_2^{\Gamma})\Sigma^{\text{val}}$, which contains only couplings that are irrelevant with respect to the Fermi-liquid RG (self-energy curvature, quasiparticle lifetime, and non-forward vertex corrections), as the perturbation. The tree level, $\delta\Sigma = 0$, gives

$$L_{\text{KS}} = \bar{\psi} (z_{\Gamma}^{\text{val}})^{-1} \left[\hat{Z}_{\text{core}}^{-1} \partial_{\tau} - (\hat{H}_{\text{KS}} - \epsilon_F) \right] \psi, \quad (10)$$

where $(\hat{Z}_{\text{core}}^{-1})_{\mathbf{G}\mathbf{G}'} = \delta_{\mathbf{G}\mathbf{G}'} + \sum_c f_{|\mathbf{k}+\mathbf{G}|}^c f_{|\mathbf{k}+\mathbf{G}'|}^c / \Delta E_c^2$ is the core renormalization operator and $\hat{H}_{\text{KS}} = -\nabla^2/2 + V_{\text{PSP}} + V_H[n] + V_{xc}^{\text{LDA}}[n]$. Beyond tree level, $\delta\Sigma$ is expanded order by order using KS propagators and subtracted vertices; Appendix E gives the Feynman rules and explicit diagrams to second order.

Projecting Eq. (10) onto a KS eigenstate gives Eq. (1). Since $\hat{Z}_{\text{core}}^{-1}$ acts only on ∂_{τ} and not on $\hat{H}_{\text{KS}}$, the KS eigenvalue equation, self-consistent density, and total energy are all unchanged from standard KS. What changes is the quasiparticle dispersion, which acquires the factor $z_{\nu\mathbf{k}}^{\text{core}}$ [Eq. (2)]. For valence electrons, the Ward identity $z^{\text{val}}\Gamma_3^{\text{val}} \approx 1$ ensures that z^{val} cancels in the pole position. No such cancellation occurs for the core: the core polarizability is suppressed by $\epsilon_F / \Delta E_c \sim 0.02\text{--}0.05$, so $\Gamma_3^{\text{core}} \approx 1$ rather than $(z^{\text{core}})^{-1}$, and $z_{\nu\mathbf{k}}^{\text{core}}$ survives in the quasiparticle energy.

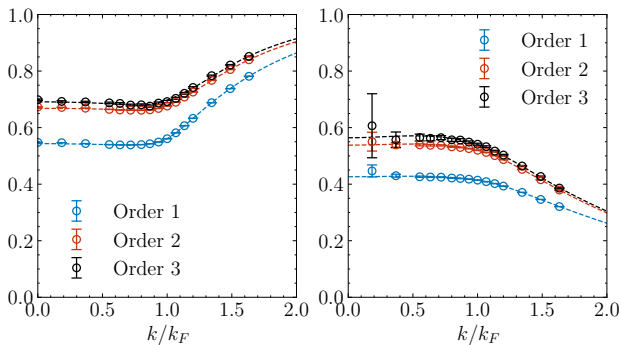


FIG. 3. Quasiparticle properties of the UEG at $r_s = 4$ from DiagMC [19]. Left: the renormalization factor $z(k)$ is nearly constant within the Fermi sea. Right: $(\text{Re } \Sigma - \mu)/k^2$ is flat, implying $m^* \approx m$. Both properties reflect an approximate Galilean invariance $\Sigma(\mathbf{k}, \omega) \approx \Sigma(\omega - k^2/2m)$.

Results.—We validate the theory across seven elements. For lithium ($Z = 3$), the hydrogenic core enables analytic evaluation of all integrals (Appendix F). The static EFT potential agrees with the conventional GTH [18] pseudopotential to $\sim 10\%$ at Γ (Appendix F), confirming that the derivation recovers known pseudopotential physics. This validates the two-step approach: a standard PSP handles the static core screening, while the EFT provides the dynamical correction. For heavier elements (Na, K, Ca, Mg, Al, Si), core orbitals are obtained numerically from a self-consistent radial KS solver (Appendix H). All standard $Z_{\text{val}} = 1$ pseudopotentials yield identical Γ -point depths to within 1 meV (Appendix H), confirming that the correction depends on atomic core properties, not on the pseudopotential used for the solid-state calculation.

The predicted QP band structures (Fig. 1, Table I) confirm the central prediction: the correction is large (20–35%) for alkali and alkaline-earth metals, where ΔE_c is only a few Hartree, and negligible ($< 5\%$) for Al and Si, where ΔE_c is several times larger. For each element, a single core s -orbital contributes $> 99\%$ of the correction; deeper channels are suppressed by their large ΔE_c . The EFT agrees with eDMFT to better than 0.15 eV for Li and K, and reproduces the ARPES bandwidth for Na, Ca, Mg, and Al. G_0W_0 narrows the bandwidth by only 4–10% (Table I), confirming that perturbative GW captures valence screening but misses the core dynamical effect. The PBE functional (dotted in Fig. 1) closely tracks LDA (solid), showing that the correction is insensitive to the exchange-correlation functional.

The trend across elements is transparent: near Γ , $1/z^{\text{core}} - 1 \approx (f_0^c)^2/\Delta E_c^2$. The form factor f_0^c is of order the Coulomb self-energy J_c of the core orbital (~ 1 Ha, nearly Z -independent), while ΔE_c grows rapidly with nuclear charge. For alkali metals the outermost core s -orbital sits just below the valence, giving ΔE_c of only a few Hartree and a large correction. Within the [Ne]-core

series, z_{Γ}^{core} increases monotonically (Na 0.80, Mg 0.91, Al 0.96, Si 0.98) as ΔE_{2s} roughly triples while f_0^{2s} stays near 1 Ha. The non-monotonic ordering among alkali metals ($K < Li < Na$) reflects different core shells: K’s 3s has the smallest ΔE_c (1.72 Ha).

The close EFT–eDMFT agreement reveals that the bandwidth narrowing in alkali and alkaline-earth metals is dominated by a single physical mechanism: the frequency dependence of the core–valence self-energy, captured by the dynamical term in Eq. (6). That a leading-order analytic formula reproduces a full quantum Monte Carlo impurity calculation indicates that higher-order corrections (multi-hole excitations, suppressed by $\epsilon_F/\Delta E_c$) are small for these systems. Even for K, the worst case in our series, $\epsilon_F/\Delta E_{3s} \approx 0.08/1.72 \approx 0.05$ remains small. For each atom, a single core s -orbital contributes $> 99\%$ of the correction; deep core orbitals are suppressed by their large ΔE_c ($\propto Z^2$), and p -orbitals vanish near Γ by angular momentum selection ($f_p \propto j_1 \rightarrow 0$).

The correction is robust to the choice of both exchange-correlation functional and pseudopotential. Replacing the LDA functional and PSP with PBE (functional and PSP together) at the same Z_{val} changes the bandwidth narrowing by < 0.2 percentage points for Mg and Al (Table II, Fig. 6). Even when Z_{val} increases from 1 to 9 (Na, K) or 2 to 10 (Ca), promoting semicore shells into the valence, the narrowing of the conduction band (the only band corrected, to avoid double counting with the now-explicit semicore states) changes by only 1–2 percentage points. This insensitivity is expected: $z_{\nu\mathbf{k}}^{\text{core}}$ depends on core orbital properties (form factors and excitation energies) computed from the isolated atom, not from the solid-state functional or pseudopotential.

Discussion.—The central message of this Letter is that the Kohn–Sham Hamiltonian, while not derivable from DFT, emerges as the tree-level output of the EFT of the electron gas: the approximate Galilean invariance of the UEG and the core–valence scale separation together force the tree-level quasiparticle dispersion to coincide

TABLE I. Occupied bandwidth (Γ -point depth in eV, measured below ϵ_F for metals and below VBM for Si). LDA, G_0W_0 , and eDMFT for Li–Mg from Ref. [9]. ARPES: Na [1, 2]; K [3]; Ca [23]; Mg [4]; Al [5]; Si [6].

	LDA	G_0W_0	eDMFT	EFT QP (LDA)	EFT QP (PBE)	Expt
Li	3.48	3.39	2.60	2.62	2.55	—
Na	3.30	3.15	2.84	2.62	2.63	2.65–2.78
K	2.15	2.00	1.42	1.49	1.44	1.60
Ca	3.98	3.79	3.24	3.06	3.19	3.30
Mg	6.89	6.66	6.18	6.31	6.34	6.15
Al	11.18	—	—	10.70	10.72	10.6
Si	12.26	—	—	11.81	11.85	12.4

with KS eigenvalues, up to the controlled correction $z_{\nu\mathbf{k}}^{\text{core}}$. This reframes the KS Hamiltonian as the leading-order output of a systematic many-body theory with a known expansion parameter $\epsilon_F/\Delta E_c$, rather than an auxiliary construction of DFT.

The EFT-KS framework also offers a natural route to the double-counting problem that plagues beyond-DFT methods. In DFT+GW, the subtraction $\Sigma_{GW} - V_{xc}$ is ad hoc because DFT provides no systematic relationship between V_{xc} and the self-energy diagrams. In DFT+DMFT, the double-counting correction Σ_{dc} is even more ambiguous, with competing prescriptions (fully localized limit, around mean field, etc.) that lack a controlled justification. These ambiguities arise from combining two incompatible languages: the wavefunction picture of DFT and the field-theoretic framework of the many-body solver. In the EFT, both share the same language: the renormalized perturbation theory defines the tree-level propagator (KS) and the residual $\delta\Sigma = (1 - P_2^\Gamma)\Sigma$ through the same projector, so what is already included at tree level and what remains as a correction is unambiguous at each order.

In this work we have applied the EFT to quasiparticle band narrowing, the simplest observable sensitive to z^{core} . The framework extends naturally in several directions. For semiconductors, the tree-level action retains an unscreened long-range Coulomb component that, once included, yields a KS Hamiltonian with hybrid-functional form, potentially providing a first-principles explanation of the band-gap problem without tuning parameters or GW-type diagrammatic calculations. The dynamical term in Eq. (6) that produces z^{core} also encodes the core polarizability; correlated fluctuations of this polarizability between neighboring atoms give rise to long-range van der Waals interactions, which are absent in standard semilocal DFT but emerge naturally from the EFT action. For strongly correlated materials with *d*- or *f*-shell cores, where multiple dynamical channels at comparable energies produce *l*-dependent quasiparticle renormalization, the EFT-KS framework may provide a natural embedding in which the impurity problem is formulated in the same field-theoretic language as the lattice theory, potentially eliminating the double-counting ambiguity that has long plagued DFT+DMFT.

[Acknowledgments]

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- [25] In the action convention used here ($g_0^{-1} = -i\omega + \hat{H}_0$, $\Sigma_c = g_c^{-1} - G_2^{-1}$), the algebra gives $\delta V_{pp} = g_0^{-1} R^{-1} g_0^{-1} - \Sigma_c - \Sigma_c G_2 \Sigma_c$ with minus signs. In the physics convention ($\mathcal{G}_0^{-1} = i\omega - \hat{H}_0$, $\Sigma_c^{(\text{phys})} = \mathcal{G}_c^{-1} - \mathcal{G}_2^{-1}$), all signs are positive. The $1/\Lambda$ expansion in Appendix B derives the correct result independently of this intermediate formula.
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Appendix A: Dual-fermion derivation of the pseudopotential

We start from the Born–Oppenheimer Lagrangian for electrons in a lattice potential $V_{\text{Lat}}(\mathbf{r})$:

$$L = \int_{\mathbf{r}\sigma} \bar{\psi}_{\mathbf{r}\sigma} \left[\partial_\tau - \frac{\nabla^2}{2} + V_{\text{Lat}} - \mu \right] \psi_{\mathbf{r}\sigma} + \frac{1}{2} \int_{\mathbf{r}\mathbf{r}'} \frac{\bar{\psi}_{\mathbf{r}\sigma} \bar{\psi}_{\mathbf{r}'\sigma'} \psi_{\mathbf{r}'\sigma'} \psi_{\mathbf{r}\sigma}}{|\mathbf{r} - \mathbf{r}'|}, \quad (\text{A1})$$

in Hartree atomic units. We introduce a reference Green’s function g_c that captures the core states, defined through a hybridization R :

$$g_c^{-1} = g_0^{-1} + R, \quad (\text{A2})$$

where $g_0^{-1} = -i\omega + \hat{H}_0$ is the bare propagator ($\hat{H}_0 = -\nabla^2/2 + V_{\text{Lat}} - \mu$). The hybridization adds a large energy $\Lambda \rightarrow \infty$ to the valence orbitals, freezing them in the reference system while keeping core states at their physical energies.

Through a Hubbard–Stratonovich transformation that decouples R and integration over the core fields [14], the effective valence propagator is:

$$g_v^{-1} = g_c^{-1} [R^{-1} - G_2] g_c^{-1}, \quad (\text{A3})$$

where G_2 is the interacting two-point function of the reference system, satisfying the Dyson equation $G_2 = g_c + g_c \Sigma_c G_2$ with core self-energy $\Sigma_c = g_c^{-1} - G_2^{-1}$. Substituting Eq. (A2) and using the Dyson equation, one obtains

$$g_v^{-1} = g_0^{-1} + \delta V_{pp}, \quad (\text{A4})$$

where the pseudopotential is identified as [25]

$$\delta V_{pp} = g_0^{-1} R^{-1} g_0^{-1} + \Sigma_c^{(\text{phys})} + \Sigma_c^{(\text{phys})} \mathcal{G}_2 \Sigma_c^{(\text{phys})}, \quad (\text{A5})$$

where $\Sigma_c^{(\text{phys})} > 0$ for Hartree (screening the nucleus). The last term enforces orthogonality to core states (it diverges for core matrix elements), while $\Sigma_c^{(\text{phys})} + \Sigma_c^{(\text{phys})} \mathcal{G}_2 \Sigma_c^{(\text{phys})}$ carries the physical core–valence interaction.

Starting from the partition function $Z = \int \mathcal{D}[\bar{\psi}, \psi] e^{-\int_0^\beta d\tau L}$, we rewrite the hybridization contribution using:

$$e^{\int_{XX'} \bar{\psi}_X R(X, X') \psi_{X'}} = \frac{1}{\det R^{-1}} \int \mathcal{D}[\bar{\phi}, \phi] e^{-\bar{\phi} R^{-1} \phi - (\bar{\phi} \psi + \bar{\psi} \phi)}. \quad (\text{A6})$$

The partition function becomes

$$Z = \frac{1}{\det R^{-1}} \int \mathcal{D}[\bar{\phi}, \phi] e^{-\bar{\phi} R^{-1} \phi} \int \mathcal{D}[\bar{\psi}, \psi] e^{-S_c[\bar{\psi}, \psi] - (\bar{\phi} \psi + \bar{\psi} \phi)}, \quad (\text{A7})$$

where S_c is the reference action. Integrating out the ψ fields yields the cumulant expansion:

$$\int \mathcal{D}[\bar{\psi}, \psi] e^{-S_c - (\bar{\phi} \psi + \bar{\psi} \phi)} = e^{\bar{\phi} G_2 \phi + O(\phi^4)}. \quad (\text{A8})$$

Redefining $\bar{\psi} = \bar{\phi} g_c$ to restore the original field normalization gives $g_v^{-1} = g_c^{-1} [R^{-1} - G_2] g_c^{-1}$. Expanding:

$$\begin{aligned} g_v^{-1} &= g_c^{-1} R^{-1} g_c^{-1} - g_c^{-1} G_2 g_c^{-1} \\ &= (g_0^{-1} + R) R^{-1} (g_0^{-1} + R) - g_c^{-1} G_2 g_c^{-1}. \end{aligned} \quad (\text{A9})$$

Using $G_2^{-1} = g_c^{-1} - \Sigma_c$ and simplifying yields Eq. (A5).

Appendix B: $1/\Lambda$ expansion and matrix elements

We assume core electrons are localized on individual ions. The reference Hamiltonian for a single ion with N_c core electrons is

$$\hat{H}_c = \sum_j E_j^c a_j^\dagger a_j + \frac{1}{2} \sum_{ijkl} v_{ijkl} a_i^\dagger a_j^\dagger a_l a_k, \quad (\text{B1})$$

where $\{j\}$ runs over hydrogen-like orbitals, $E_j^c = E_j$ for core orbitals, $E_j^c = E_0 + \Lambda$ for valence orbitals ($\Lambda \rightarrow \infty$), and $v_{ijkl} = \langle ij || kl \rangle$ are antisymmetrized two-electron integrals.

The ground state of the reference system is

$$|\Phi_0\rangle = |N_c, \xi_0\rangle + \sum_{\xi_1} \frac{\langle N_c, \xi_0 || N_c, \xi_1 \rangle}{n(\xi_1) \Lambda} |N_c, \xi_1\rangle + O(1/\Lambda^2), \quad (\text{B2})$$

where $|N_c, \xi_0\rangle$ is the Slater determinant with all core orbitals filled and $n(\xi_1)$ counts valence orbitals in $|N_c, \xi_1\rangle$.

We work in the hydrogen orbital basis, labeling core orbitals as i, j and valence orbitals as s, t . The interacting Green’s function decomposes as $(G_2)_{st} = (G_2^p)_{st} + (G_2^h)_{st}$.

Particle part.—Expanding $N_c + 1$ -particle states to $O(1/\Lambda^2)$:

$$(G_2^p)_{st} = \frac{\delta_{st}}{\Lambda} + \frac{i\omega \delta_{st} - \Sigma_{st}^{\text{HF}}}{\Lambda^2} + O(1/\Lambda^3), \quad (\text{B3})$$

where $\Sigma_{st}^{\text{HF}} = \sum_c n_c [\langle cs || ct \rangle_{\text{Hartree}} - \langle cs || tc \rangle_{\text{exchange}}]$.

Hole part.—The ground state has $O(1/\Lambda)$ admixture:

$$(G_2^h)_{st} = -\frac{1}{\Lambda^2} \sum_c \frac{M_{sc} M_{tc}^*}{i\omega + (E_c - E_0)}, \quad (\text{B4})$$

where $M_{sc} = \langle N_c, \xi_0 | N_c, \xi_1^{sc} \rangle$.

Combining and taking $\Lambda \rightarrow \infty$:

$$(\delta V_{pp})_{st} = \Sigma_{st}^{\text{HF}} + \sum_c \frac{M_{sc} M_{tc}^*}{i\omega + (E_c - E_0)}, \quad (\text{B5})$$

which is Eq. (6). The $g_0^{-1} R^{-1} g_0^{-1}$ term gives $(g_v^{-1})_{11} =$

∞ , projecting out core states.

Appendix C: Four-term pseudopotential in coordinate space

Evaluating Eq. (6) in coordinate space for a core of spin-paired orbitals $\{\phi_c\}$ with radial functions $u_c(r) = rR_c(r)$:

$$\begin{aligned} \delta V_{pp}^{(\text{ion})}(\mathbf{r}, \mathbf{r}') = & \underbrace{\sum_c n_c V_{H,c}(r) \delta(\mathbf{r} - \mathbf{r}')}_{\text{Term 1: Hartree}} - \underbrace{\sum_c n_c \phi_c^*(\mathbf{r}) \phi_c(\mathbf{r}')}_{\text{Term 2: Exchange}} \frac{1}{|\mathbf{r} - \mathbf{r}'|} \\ & - \underbrace{\sum_c n_c \phi_c^*(\mathbf{r}) \phi_c(\mathbf{r}') [V_{H,c}(r) + V_{H,c}(r') - 2J_c]}_{\text{Term 3: Projection (static } \Sigma G_2 \Sigma)} + \underbrace{\sum_c n_c \phi_c^*(\mathbf{r}) \phi_c(\mathbf{r}') \frac{(V_{H,c}(r) - J_c)(V_{H,c}(r') - J_c)}{i\omega + \Delta E_c}}_{\text{Term 4: Dynamical}}, \end{aligned} \quad (\text{C1})$$

where n_c is the occupation, $V_{H,c}(r) = r^{-1} \int_0^r u_c^2 dr' + \int_r^\infty u_c^2 / r' dr'$ is the Hartree potential of orbital c , $J_c = \int u_c^2 V_{H,c} dr$ is the self-Coulomb integral, and $\Delta E_c = E_{\text{core}}^{(\text{hole } c)} - E_{\text{core}}^{(\text{gs})}$ is the core excitation energy.

Term 1 is the local Hartree potential, screening the nuclear charge. **Term 2** is the nonlocal Fock exchange, attractive and l -dependent. **Term 3** enforces core-valence orthogonality. **Term 4** is the dynamical correction absent in conventional pseudopotentials.

Appendix D: Derivation of the quasiparticle formula

We derive Eq. (2) from the Dyson equation, showing that it is exact for the separable self-energy of Eq. (6) up to corrections of order $(\epsilon_{\text{KS}} - \epsilon_F)^2 / \Delta E^2 \sim 0.2\%$.

Static-dynamic splitting

The EFT pseudopotential splits into static and dynamic parts: $V_{\text{pp}}(\mathbf{G}, \mathbf{G}'; \omega) = V_{\text{stat}}(\mathbf{G}, \mathbf{G}') + \Sigma_{\text{dyn}}(\mathbf{G}, \mathbf{G}'; \omega)$. The static part is a conventional pseudopotential (local + KB projectors). The dynamic part has poles at $\omega = -\Delta E_c$ ($\Delta E_c > 0$) and is separable:

$$\Sigma_{\text{dyn}}(\mathbf{G}, \mathbf{G}'; \omega) = \sum_c \frac{f_{|\mathbf{k}+\mathbf{G}|}^c f_{|\mathbf{k}+\mathbf{G}'|}^c}{\omega + \Delta E_c}. \quad (\text{D1})$$

KS basis and self-energy projection

Standard DFT diagonalizes the static KS Hamiltonian $H_{\text{KS}} = T + V_{\text{stat}} + V_H + V_{\text{xc}}$, giving eigenstates $|\psi_{\nu\mathbf{k}}\rangle$ with eigenvalues $\varepsilon_{\nu\mathbf{k}}$ and Bloch coefficients $c_{\nu\mathbf{k}}(\mathbf{G})$. Projecting Σ_{dyn} onto this basis, define

$$F_{\nu\mathbf{k}}^c = \sum_{\mathbf{G}} c_{\nu\mathbf{k}}(\mathbf{G}) f_{|\mathbf{k}+\mathbf{G}|}^c. \quad (\text{D2})$$

The self-energy matrix in the KS basis is

$$\Sigma_{\nu\nu'}(\mathbf{k}; \omega) = \sum_c \frac{F_{\nu}^c (F_{\nu'}^c)^*}{\omega + \Delta E_c}, \quad (\text{D3})$$

which is *not* diagonal in band index ν .

Double counting and subtraction

The full Dyson equation $\det[\omega \delta_{\nu\nu'} - \varepsilon_{\nu\mathbf{k}} \delta_{\nu\nu'} - \Sigma_{\nu\nu'}(\mathbf{k}; \omega)] = 0$ gives unphysical results because Σ_{dyn} contains a static piece:

$$\Sigma_{\text{dyn}}(\omega) = \underbrace{\frac{|f\rangle\langle f|}{\varepsilon_F + \Delta E_c}}_{\Sigma_0} + \underbrace{\frac{|f\rangle\langle f| (\varepsilon_F - \omega)}{(\omega + \Delta E_c)(\varepsilon_F + \Delta E_c)}}_{\delta\Sigma(\omega)}. \quad (\text{D4})$$

The static part $\Sigma_0 \sim 20$ eV (for Li) is already included in H_{KS} via V_{xc} and the pseudopotential. Including it again double-counts.

The subtracted diagonal self-energy is

$$\Sigma_{\nu\nu}(\omega) - \Sigma_{\nu\nu}(\varepsilon_F) = -\frac{|F_\nu^c|^2 (\omega - \varepsilon_F)}{(\omega + \Delta E_c)(\varepsilon_F + \Delta E_c)}. \quad (\text{D5})$$

Exact solution and z -factor

Setting $\delta = \omega - \varepsilon_F$ and $s = \varepsilon_{\nu\mathbf{k}} - \varepsilon_F$, with $D = \varepsilon_F + \Delta E_c$:

$$\delta = \frac{s D(\delta + D)}{D(\delta + D) + |F_\nu^c|^2}. \quad (\text{D6})$$

Since $|\delta| \sim 0.1 \text{ Ha} \ll D \sim 2.7 \text{ Ha}$, setting $\delta + D \approx D$ gives

$$\delta \approx \frac{s}{1 + |F_\nu^c|^2/D^2} = z_{\nu\mathbf{k}}^{\text{core}} (\varepsilon_{\nu\mathbf{k}} - \varepsilon_F), \quad (\text{D7})$$

where $z_{\nu\mathbf{k}}^{\text{core}} = 1/(1 + |F_\nu^c|^2/D^2) \approx 1/(1 + |F_\nu^c|^2/\Delta E_c^2)$, recovering Eq. (2) with $z_{\nu\mathbf{k}}^{\text{core}} = 1/(1 + \sum_c |F_{\nu\mathbf{k}}^c|^2/\Delta E_c^2)$. No linearization in ω was used; the only approximation is $\delta/D \sim 4\%$.

Off-diagonal (band mixing) is suppressed

The subtracted off-diagonal matrix elements $\Sigma_{\nu\nu'}^{\text{sub}}(\omega) = -F_\nu^c(F_{\nu'}^c)^* (\omega - \varepsilon_F)/[(\omega + \Delta E_c)D]$ vanish as $\omega \rightarrow \varepsilon_F$. The second-order correction from band mixing is $\delta\varepsilon_\nu^{(2)} \sim |F_\nu^c|^2 |F_{\nu'}^c|^2 \delta^2/[D^4 (\varepsilon_\nu - \varepsilon_{\nu'})] = O(\delta^2/D^2) \sim 0.2\%$, making the diagonal approximation excellent.

Appendix E: Renormalized perturbation theory to second order

This appendix spells out the renormalized perturbation theory (RPT) used in the main text. The purpose of the tree-level Lagrangian in Eq. (10) is to define the zeroth-order propagator. The residual self-energy $\delta\Sigma$ is then computed from ordinary valence interaction diagrams, but with the relevant local shift, the representative Γ -point residue, and the static charge-vertex matching already included in g^{KS} subtracted order by order.

All formulas below are finite-temperature Matsubara formulas. Fermion frequencies are $\omega_n = (2n+1)\pi/\beta$ and boson frequencies are $\Omega_\ell = 2\pi\ell/\beta$. We use the same $\nu\mathbf{G}$ language as in the main text. A KS Bloch state is

$$\psi_{\nu\mathbf{k}\sigma}(\mathbf{r}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{G}} c_{\nu\mathbf{k}}(\mathbf{G}) e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} \chi_\sigma, \quad (\text{E1})$$

and the tree-level Matsubara propagator is

$$g_{\nu}^{\text{KS}}(\mathbf{k}, i\omega_n) = \frac{R_{\nu\mathbf{k}}}{i\omega_n - E_{\nu\mathbf{k}}^{(0)}}, \quad E_{\nu\mathbf{k}}^{(0)} = z_{\nu\mathbf{k}}^{\text{core}} (\epsilon_{\nu\mathbf{k}}^{\text{KS}} - \epsilon_F), \quad (\text{E2})$$

with $R_{\nu\mathbf{k}} = z^{\text{val}} z_{\nu\mathbf{k}}^{\text{core}}$ in the diagonal approximation. In the plane-wave representation this is

$$g_{\mathbf{G}\mathbf{G}'}^{\text{KS}}(\mathbf{k}, i\omega_n) = \sum_{\nu} c_{\nu\mathbf{k}}(\mathbf{G}) g_{\nu}^{\text{KS}}(\mathbf{k}, i\omega_n) c_{\nu\mathbf{k}}^*(\mathbf{G}'). \quad (\text{E3})$$

Keeping $\hat{Z}_{\text{core}}^{-1}$ as a matrix gives the same rules with $g_{\mathbf{G}\mathbf{G}'}^{\text{KS}}$ used directly.

The valence interaction is the core-dressed Coulomb vertex v^{val} . Its finite-temperature action is written in terms of Matsubara density vertices,

$$S_{\text{int}}^{\text{val}} = \frac{1}{2\beta\mathcal{V}} \sum_{\mathbf{q}\mathbf{Q}\Omega_\ell} v_{\mathbf{q}+\mathbf{Q}}^{\text{val}} \rho_{\mathbf{Q}}(\mathbf{q}, i\Omega_\ell) \rho_{-\mathbf{Q}}(-\mathbf{q}, -i\Omega_\ell), \quad (\text{E4})$$

where

$$\begin{aligned} \rho_{\mathbf{Q}}(\mathbf{q}, i\Omega_\ell) &= \sum_{\substack{\mathbf{k}, \omega_n, \sigma \\ \nu\nu'}} \Lambda_{\nu\nu'}^{\mathbf{Q}}(\mathbf{k} + \mathbf{q}, \mathbf{k}) \bar{\psi}_{\nu, \mathbf{k}+\mathbf{q}, \sigma}(i\omega_n + i\Omega_\ell) \\ &\quad \times \psi_{\nu', \mathbf{k}, \sigma}(i\omega_n), \end{aligned} \quad (\text{E5})$$

with the $\nu\mathbf{G}$ density matrix element

$$\Lambda_{\nu\nu'}^{\mathbf{Q}}(\mathbf{k} + \mathbf{q}, \mathbf{k}) = \sum_{\mathbf{G}} c_{\nu, \mathbf{k}+\mathbf{q}}^*(\mathbf{G} + \mathbf{Q}) c_{\nu'\mathbf{k}}(\mathbf{G}). \quad (\text{E6})$$

Equivalently, for external labels $1 = (\nu_1, \mathbf{k}_1, \sigma_1)$ etc., the static Coulomb vertex is

$$\begin{aligned} V_{12;34}^{\text{val}} &= \frac{\delta_{\mathbf{k}_1+\mathbf{k}_2, \mathbf{k}_3+\mathbf{k}_4}}{\mathcal{V}} \sum_{\mathbf{Q}} v_{\mathbf{k}_1-\mathbf{k}_3+\mathbf{Q}}^{\text{val}} \\ &\quad \times \Lambda_{\nu_1\nu_3}^{\mathbf{Q}}(\mathbf{k}_1, \mathbf{k}_3) \Lambda_{\nu_2\nu_4}^{-\mathbf{Q}}(\mathbf{k}_2, \mathbf{k}_4) \delta_{\sigma_1\sigma_3} \delta_{\sigma_2\sigma_4}, \end{aligned} \quad (\text{E7})$$

with momentum conservation understood modulo a reciprocal lattice vector. We define the antisymmetrized Coulomb vertex $\bar{V}_{12;34}^{\text{val}} = V_{12;34}^{\text{val}} - V_{12;43}^{\text{val}}$. The core dressing changes only short-distance Wilson coefficients and high-frequency structure at the valence scale: the core polarizability is suppressed by $\epsilon_F/\Delta E_c$, so within the valence Fermi ball $v_{\mathbf{q}}^{\text{val}} = 4\pi/q^2 + O(\epsilon_F/\Delta E_c)$ for $\mathbf{q} \neq 0$, up to short-range pieces. The $\mathbf{q} = 0$ Hartree component is fixed by charge neutrality and, in the RPT, is part of the matched static term $V_H[n]$ that is subtracted by the two-point projector P_2^Γ below. Since the interaction remains long ranged, the expansion in powers of $v_{\mathbf{q}}^{\text{val}}$ is a formal RPT; in an infrared-sensitive metallic calculation one may further reorganize the same series by resumming ring diagrams into a screened interaction W^{val} , without changing the matching conditions described here. For the charge-vertex counterterm it is useful to display the endpoint correction in the same $\nu\mathbf{G}$ density-vertex notation:

$$\Lambda_{3,\nu\nu'}^{\mathbf{Q},n}(\mathbf{k} + \mathbf{q}, \mathbf{k}) = \sum_{\mu\mu'} [\mathcal{C}_3^{(n)}(\mathbf{q}, 0)]_{\nu\nu';\mu\mu'} \Lambda_{\mu\mu'}^{\mathbf{Q}}(\mathbf{k} + \mathbf{q}, \mathbf{k}). \quad (\text{E8})$$

The endpoint-corrected Coulomb vertex entering second-order self-energy diagrams is therefore

$$\bar{V}_{12;34}^{(3,n)} = \frac{\delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4}}{\mathcal{V}} \sum_{\mathbf{Q}} v_{\mathbf{k}_1-\mathbf{k}_3+\mathbf{Q}}^{\text{val}} \left[\Lambda_{3,\nu_1\nu_3}^{\mathbf{Q},n}(\mathbf{k}_1, \mathbf{k}_3) \Lambda_{\nu_2\nu_4}^{-\mathbf{Q}}(\mathbf{k}_2, \mathbf{k}_4) + \Lambda_{\nu_1\nu_3}^{\mathbf{Q}}(\mathbf{k}_1, \mathbf{k}_3) \Lambda_{3,\nu_2\nu_4}^{-\mathbf{Q},n}(\mathbf{k}_2, \mathbf{k}_4) \right]_{\text{AS}}, \quad (\text{E9})$$

where AS denotes the same exchange antisymmetrization as in $\bar{V}^{\text{val}}$. Only the P_3^0 component of $\mathcal{C}_3$ is fixed by the matching condition; the remaining momentum- and frequency-dependent charge-vertex structure is part of the residual EFT expansion.

The RPT therefore contains two counterterms. A two-point counterterm $\mathcal{C}_2 = \sum_n \mathcal{C}_2^{(n)}$ subtracts the matched self-energy data, and a three-point counterterm $\mathcal{C}_3 = \sum_n \mathcal{C}_3^{(n)}$ subtracts the matched charge vertex:

$$\mathcal{C}_2^{(n)} = -P_2^\Gamma \Sigma_{\text{eff}}^{(n)}, \quad \mathcal{C}_3^{(n)} = -P_3^0 \Gamma_{3,\text{eff}}^{(n)}. \quad (\text{E10})$$

For the matching convention used in the main text these are two different projections. The two-point projection P_2^Γ is a local density projection plus a single Γ -point residue matching. In plane-wave representation,

$$(P_2^\Gamma \Sigma)_{\mathbf{G}\mathbf{G}'}(\mathbf{k}, i\omega) = [\delta\mu_\Sigma(n)]_{\mathbf{G}-\mathbf{G}'} + \delta_{\mathbf{G}\mathbf{G}'} A_\Gamma(i\omega - \xi_{\mathbf{k}+\mathbf{G}}), \quad (\text{E11})$$

where

$$\delta\mu_\Sigma(n) = \Sigma^{\text{UEG}}(n; k_F, i0^+), \quad A_\Gamma = \left. \frac{\partial \Sigma_\Gamma^{\text{val}}(i\omega)}{\partial(i\omega)} \right|_{\omega=0} = 1 - (z_\Gamma^{\text{val}})^{-1}. \quad (\text{E12})$$

The first coefficient gives the local scalar density shift, whose leading matched value is $V_H + V_{xc}^{\text{LDA}}$; the second coefficient fixes the representative residue z_Γ^{val} used in the main text. The effective mass is not independently fitted: the Galilean-invariant linear combination $i\omega - \xi_{\mathbf{k}+\mathbf{G}}$ is used, and the residual deviation from $m^* = m$ is left in $\delta\Sigma_{\text{curv}}$.

For the static density three-point vertex we use a different projection, P_3^0 , which takes the s-wave component of the long-wavelength, zero-frequency charge vertex on the Fermi surface,

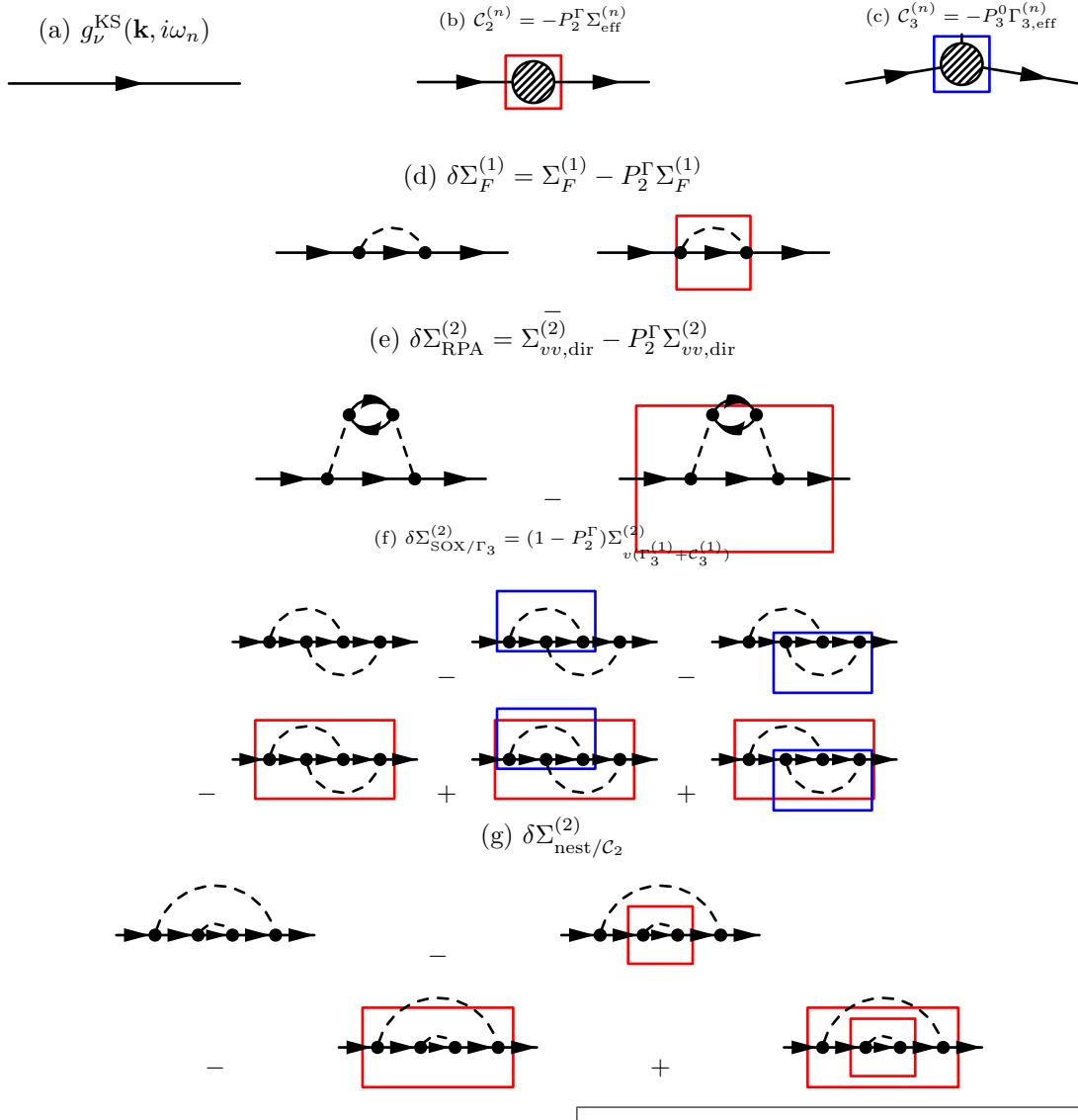
$$P_3^0 \Gamma_3 = \left\langle \lim_{\mathbf{q} \rightarrow 0} \lim_{\Omega \rightarrow 0} \Gamma_3(\mathbf{k}, i0^+; \mathbf{q}, i\Omega) \right\rangle_{\text{FS}}, \quad (\text{E13})$$

with the renormalization condition $z_\Gamma^{\text{val}} P_3^0 \Gamma_3^{\text{val}} = 1$. Here $\langle \dots \rangle_{\text{FS}}$ denotes the normalized Fermi-surface average; the superscript 0 labels the s-wave charge channel, not the Γ point. This is the charge/number Ward identity used in the main text. The two projectors are tied only by this matching condition and the observed near flatness of $z(k)$. Order by order,

$$\delta\Sigma^{(n)} = \Sigma_{\text{eff}}^{(n)} + \mathcal{C}_2^{(n)} = (1 - P_2^\Gamma) \Sigma_{\text{eff}}^{(n)}, \quad (\text{E14})$$

$$\delta\Sigma = \delta\Sigma^{(1)} + \delta\Sigma^{(2)} + \dots$$

Here $\Sigma_{\text{eff}}^{(n)}$ means all one-particle-irreducible self-energy diagrams of total order n built from Coulomb vertices and lower-order two- and three-point counterterm insertions, using g^{KS} lines. A first-order $\mathcal{C}_3$ insertion on a Coulomb endpoint therefore contributes to the second-order self-energy. This is the subtracted-diagram meaning of the residual self-energy expansion. Diagrammatically, the drawings below are finite-temperature Matsubara contraction topologies. Each internal fermion line carries $(\nu_i, \mathbf{k}_i, i\omega_i)$ and contributes a factor $\beta^{-1} \sum_{\omega_i} \sum_{\nu_i \mathbf{k}_i}$; each dashed Coulomb line carries $(\mathbf{q}, i\Omega_\ell)$ and the static core-dressed interaction $v_{\mathbf{q}+\mathbf{Q}}^{\text{val}}$. The actual algebra is the Matsubara expansion written after the figure.



Panels (a)–(c) are the elementary renormalized line and the two projected counterterm insertions. A colored projector box in a counterterm panel is not a new bare vertex: it marks the corresponding self-energy or charge-vertex subdiagram after applying the matching projector. Red boxes denote P_2^Γ projections of self-energy subdiagrams, while blue boxes denote P_3^0 projections of charge vertices, e.g., $\mathcal{C}_2^{(1)} = -P_2^\Gamma \Sigma_{\text{eff}}^{(1)}$ and $\mathcal{C}_2^{(2)} = -P_2^\Gamma \Sigma_{\text{eff}}^{(2)}$. Panel (d) is the first-order residual self-energy $\delta \Sigma^{(1)} = (1 - P_2^\Gamma) \Sigma_F^{(1)}$: the ordinary Fock diagram and its projected counterterm. The Hartree tadpole is not drawn because it is exactly the KS Hartree potential in the matched density channel, so $\Sigma_H^{(1)} - P_2^\Gamma \Sigma_H^{(1)} = 0$. Panel (e) is the direct/RPA second-order residual pair: the polarization-bubble self-energy and its red-box projected counterdiagram. Panel (f) is the SOX/charge-vertex residual sector. Its unboxed graph is the expanded crossed/SOX self-energy contained in $\Sigma_{vv}^{(2)}$. The

two blue-box graphs subtract the two possible $P_3^0 \Gamma_{3,\text{eff}}^{(1)}$ charge-vertex subdiagrams, and the second row applies the outer P_2^Γ self-energy projection, producing the red-only and red-over-blue counterdiagrams. A second-order $\mathcal{C}_3^{(2)}$ counterterm belongs to the three-point vertex function itself and contributes to the self-energy only at the next order through an additional Coulomb line. Panel (g) is the nested-Fock sector. The first graph is the nested-Fock part of $\Sigma_{vv}^{(2)}$, whose inner two-point subdiagram is a first-order Fock self-energy on the internal fermion line. The second graph inserts its P_2^Γ subtraction, i.e. the $\mathcal{C}_2^{(1)}$ counterterm contribution $\Sigma_{v\mathcal{C}_2}^{(2)}$, and the second row applies the outer P_2^Γ projection to both terms. The residual diagrams are obtained by adding these counterterm diagrams to the ordinary diagrams so that the matched two-point and three-point data are not counted twice.

At first order the Coulomb contraction before the KS

subtraction is the finite-temperature Hartree–Fock term,

$$\begin{aligned}\Sigma_{\text{eff},\nu\nu'}^{(1)}(\mathbf{k}) &\equiv \Sigma_{v,\nu\nu'}^{(1)}(\mathbf{k}) \\ &= \frac{1}{\beta} \sum_{\mu\mathbf{p}\omega_m} \bar{V}_{\nu\mu;\nu'\mu}^{\text{val}}(\mathbf{k},\mathbf{p};\mathbf{k},\mathbf{p}) g_{\mu}^{\text{KS}}(\mathbf{p},i\omega_m) \\ &= \sum_{\mu\mathbf{p}} n_{\mu\mathbf{p}} \bar{V}_{\nu\mu;\nu'\mu}^{\text{val}}(\mathbf{k},\mathbf{p};\mathbf{k},\mathbf{p}),\end{aligned}\tag{E15}$$

where $n_{\mu\mathbf{p}} = f_{\beta}(E_{\mu\mathbf{p}}^{(0)})$ is the finite-temperature Fermi function and spin sums are suppressed. We split $\Sigma_v^{(1)} = \Sigma_H^{(1)} + \Sigma_F^{(1)}$. Because the reference propagator already contains the KS Hartree potential, $P_2^{\Gamma}\Sigma_H^{(1)} = \Sigma_H^{(1)}$ and

the Hartree residual vanishes. The first-order two-point counterterm and residual are

$$\begin{aligned}\mathcal{C}_{2,\nu\nu'}^{(1)}(\mathbf{k}) &= -[P_2^{\Gamma}\Sigma_{\text{eff}}^{(1)}]_{\nu\nu'}(\mathbf{k}), \\ \delta\Sigma_{\nu\nu'}^{(1)}(\mathbf{k}) &= [(1 - P_2^{\Gamma})\Sigma_F^{(1)}]_{\nu\nu'}(\mathbf{k}).\end{aligned}\tag{E16}$$

Thus neither the large local Hartree shift nor the LDA exchange-correlation Wilson coefficient is counted again in perturbation theory.

At second order the ordinary finite-temperature two-Coulomb self-energy is the sum of the direct bubble and vertex/SOX topologies, written compactly as

$$\begin{aligned}\Sigma_{vv,\nu\nu'}^{(2)}(\mathbf{k},i\omega_n) &= \frac{1}{2\beta^2} \sum_{\mu\lambda\eta} \sum_{\mathbf{p}\mathbf{q}} \sum_{\omega_m,\omega_l} \bar{V}_{\nu\mu;\lambda\eta}^{\text{val}}(\mathbf{k},\mathbf{p};\mathbf{q},\mathbf{k}+\mathbf{p}-\mathbf{q}) \\ &\times \bar{V}_{\lambda\eta;\nu'\mu}^{\text{val}}(\mathbf{q},\mathbf{k}+\mathbf{p}-\mathbf{q};\mathbf{k},\mathbf{p}) \\ &\times g_{\mu}^{\text{KS}}(\mathbf{p},i\omega_m) g_{\lambda}^{\text{KS}}(\mathbf{q},i\omega_l) \\ &\times g_{\eta}^{\text{KS}}(\mathbf{k}+\mathbf{p}-\mathbf{q},i\omega_n+i\omega_m-i\omega_l).\end{aligned}\tag{E17}$$

The antisymmetrized product $\bar{V}_{\nu\mu;\lambda\eta}^{\text{val}} \bar{V}_{\lambda\eta;\nu'\mu}^{\text{val}}$ contains the unprojected RPA/direct topology in panel (e) and the SOX topology in panel (f) in the diagrammatic list above. Panel (g) singles out the nested-Fock topology, which is part of $\Sigma_{vv}^{(2)}$, because it contains a first-order Fock self-

energy subdiagram that is subtracted by $\mathcal{C}_2^{(1)}$. The corresponding mixed diagram with one Coulomb vertex and one first-order two-point counterterm insertion is

$$\begin{aligned}\Sigma_{v\mathcal{C}_2,\nu\nu'}^{(2)}(\mathbf{k},i\omega_n) &= \frac{1}{\beta} \sum_{\mu\mathbf{p}} \sum_{\omega_m} \bar{V}_{\nu\mu;\nu'\lambda}^{\text{val}}(\mathbf{k},\mathbf{p};\mathbf{k},\mathbf{p}) g_{\mu}^{\text{KS}}(\mathbf{p},i\omega_m) \\ &\times \mathcal{C}_{2,\mu\lambda}^{(1)}(\mathbf{p},i\omega_m) g_{\lambda}^{\text{KS}}(\mathbf{p},i\omega_m).\end{aligned}\tag{E18}$$

The mixed diagrams with one Coulomb vertex and one first-order charge-vertex counterterm are

$$\begin{aligned}\Sigma_{v\mathcal{C}_3,\nu\nu'}^{(2)}(\mathbf{k}) &= \frac{1}{\beta} \sum_{\mu\mathbf{p}\omega_m} \bar{V}_{\nu\mu;\nu'\mu}^{(3,1)}(\mathbf{k},\mathbf{p};\mathbf{k},\mathbf{p}) g_{\mu}^{\text{KS}}(\mathbf{p},i\omega_m) \\ &= \sum_{\mu\mathbf{p}} n_{\mu\mathbf{p}} \bar{V}_{\nu\mu;\nu'\mu}^{(3,1)}(\mathbf{k},\mathbf{p};\mathbf{k},\mathbf{p}).\end{aligned}\tag{E19}$$

Equation (E9) includes the two endpoint insertions shown by the blue boxes in panel (f). This is the self-energy counterpart of renormalizing the static charge vertex in the main text. Thus

$$\Sigma_{\text{eff}}^{(2)} = \Sigma_{vv}^{(2)} + \Sigma_{v\mathcal{C}_2}^{(2)} + \Sigma_{v\mathcal{C}_3}^{(2)}.\tag{E20}$$

Equivalently, after carrying out the Matsubara sums in $\Sigma_{vv}^{(2)}$ in the diagonal quasiparticle approximation while keeping the finite-temperature Fermi factors,

$$\begin{aligned}
\Sigma_{vv,\nu\nu'}^{(2)}(\mathbf{k}, i\omega_n) &= \frac{1}{2} \sum_{\mu\lambda\eta} \sum_{\mathbf{p}\mathbf{q}} \bar{V}_{\nu\mu;\lambda\eta}^{\text{val}}(\mathbf{k}, \mathbf{p}; \mathbf{q}, \mathbf{k} + \mathbf{p} - \mathbf{q}) \\
&\times \bar{V}_{\lambda\eta;\nu'\mu}^{\text{val}}(\mathbf{q}, \mathbf{k} + \mathbf{p} - \mathbf{q}; \mathbf{k}, \mathbf{p}) \\
&\times R_{\mu\mathbf{p}} R_{\lambda\mathbf{q}} R_{\eta, \mathbf{k} + \mathbf{p} - \mathbf{q}} \frac{n_{\mu\mathbf{p}}(1 - n_{\lambda\mathbf{q}})(1 - n_{\eta, \mathbf{k} + \mathbf{p} - \mathbf{q}}) + (1 - n_{\mu\mathbf{p}})n_{\lambda\mathbf{q}}n_{\eta, \mathbf{k} + \mathbf{p} - \mathbf{q}}}{i\omega_n + E_{\mu\mathbf{p}}^{(0)} - E_{\lambda\mathbf{q}}^{(0)} - E_{\eta, \mathbf{k} + \mathbf{p} - \mathbf{q}}^{(0)}}.
\end{aligned} \tag{E21}$$

The two terms in the numerator are the finite-temperature occupation factors for the two possible intermediate-state orientations. Direct and exchange second-order diagrams are both contained in the anti-symmetrized product of $\bar{V}^{\text{val}}$ vertices.

The second-order two-point counterterm and residual are

$$\begin{aligned}
\mathcal{C}_{2,\nu\nu'}^{(2)}(\mathbf{k}, i\omega_n) &= -[P_2^\Gamma \Sigma_{\text{eff}}^{(2)}]_{\nu\nu'}(\mathbf{k}, i\omega_n), \\
\delta\Sigma_{\nu\nu'}^{(2)}(\mathbf{k}, i\omega_n) &= \Sigma_{\text{eff},\nu\nu'}^{(2)}(\mathbf{k}, i\omega_n) + \mathcal{C}_{2,\nu\nu'}^{(2)}(\mathbf{k}, i\omega_n).
\end{aligned} \tag{E22}$$

In plane-wave language, the two-point projected piece has the form

$$\begin{aligned}
(P_2^\Gamma \Sigma_{\text{eff}}^{(n)})_{\mathbf{G}\mathbf{G}'}(\mathbf{k}, i\omega) &= [\delta V_{xc}^{(n)}[n]]_{\mathbf{G}-\mathbf{G}'} \\
&+ \delta_{\mathbf{G}\mathbf{G}'} \delta A_{\Gamma,n}(i\omega - \xi_{\mathbf{k}+\mathbf{G}}),
\end{aligned} \tag{E23}$$

where $\delta V_{xc}^{(n)}$ is the local density matching coefficient at order n , and $\delta A_{\Gamma,n}$ is fixed by the chosen representative Γ -point residue matching condition. The three-point projector is instead applied to the charge vertex,

$$P_3^0 \Gamma_{3,\text{eff}}^{(n)} = \left\langle \lim_{\mathbf{q} \rightarrow 0} \lim_{\Omega \rightarrow 0} \Gamma_{3,\text{eff}}^{(n)}(\mathbf{k}, i0^+; \mathbf{q}, i\Omega) \right\rangle_{\text{FS}}, \tag{E24}$$

and the counterterm $\mathcal{C}_3^{(n)}$ enforces the scalar Ward identity order by order. Since m^* is not independently fitted, any part not captured by the Galilean-invariant combination $i\omega - \xi_{\mathbf{k}+\mathbf{G}}$ remains in $\delta\Sigma_{\text{curv}}$.

The tree-level approximation used in the main text sets

$$\delta\Sigma^{(1)} + \delta\Sigma^{(2)} + \dots \approx 0. \tag{E25}$$

This does not mean the Coulomb interaction is ignored. Its relevant static part, the leading quasiparticle residue, and the Ward-consistent scalar vertex have already been matched into g^{KS} ; the discarded terms are only the residual gradient, nonlocal, curvature, and vertex corrections beyond that matching.

Appendix F: Lithium: explicit EFT potential

Lithium ($Z = 3$) has two 1s core electrons and one 2s valence electron. The 1s orbital is $\psi_{1s}(\mathbf{r}) = (\alpha^3/\pi)^{1/2} e^{-\alpha r}$ with $\alpha = Z - 5/16 = 2.6875$ Bohr⁻¹ (variational HF for He-like Li⁺, accounting for core-core Coulomb screening).

The relevant atomic quantities are: $\varepsilon_{1s} = \alpha^2/2 - Z\alpha = -4.449$ Ha, the self-Coulomb integral $J = 5\alpha/8 = 1.680$ Ha, and the Li⁺ ground-state energy $E_0 = 2\varepsilon_{1s} + J = -7.219$ Ha. The Coulomb potential from one 1s electron is

$$u(r) \equiv V_{H,1s}(r) = \frac{1 - e^{-2\alpha r}}{r} - \alpha e^{-2\alpha r}, \tag{F1}$$

with $u(0) = \alpha$ and $u(r) \rightarrow 1/r$ at large r .

Substituting the hydrogenic 1s into Eq. (C1):

$$\begin{aligned}
\delta V_{pp}^{(\text{ion})}(\mathbf{r}, \mathbf{r}') &= 2u(r) \delta(\mathbf{r} - \mathbf{r}') - \frac{\alpha^3}{\pi} \frac{e^{-\alpha(r+r')}}{|\mathbf{r} - \mathbf{r}'|} \\
&- \frac{\alpha^3}{\pi} e^{-\alpha(r+r')} [u(r) + u(r') - J] \\
&+ \frac{\alpha^3}{\pi} e^{-\alpha(r+r')} \frac{(u(r) - J)(u(r') - J)}{i\omega + (E_1 - E_0)}.
\end{aligned} \tag{F2}$$

Term-by-term evaluation

For Li, $N_c = 2$ (two 1s electrons, spin up and down).

Term 1: Hartree.— $V_H = 2u(r) \delta(\mathbf{r} - \mathbf{r}')$, where the factor 2 accounts for both spins.

Term 2: Exchange.—The Fock exchange (like-spin only, times 2 channels): $-(\alpha^3/\pi) e^{-\alpha(r+r')}/|\mathbf{r} - \mathbf{r}'|$.

Term 3: Projection.—The static $\Sigma_c G_2 \Sigma_c$ evaluates to: $-(\alpha^3/\pi) e^{-\alpha(r+r')} [u(r) + u(r') - J]$.

Term 4: Dynamical.—The frequency-dependent piece: $+(\alpha^3/\pi) e^{-\alpha(r+r')} (u(r) - J)(u(r') - J)/(i\omega_n + E_1 - E_0)$, where $E_1 - E_0 = 2.770$ Ha.

Local (LDA) reduction

Integrating the static terms over $\mathbf{r}'$ gives

$$V_{\text{loc}}^{(\text{static})}(r) = 2u(r) - 8e^{-\alpha r} \left[u(r) + u\left(\frac{r}{2}\right) - \frac{1}{18} u\left(\frac{3r}{2}\right) - \frac{\alpha}{27} - J \right]. \tag{F3}$$

The total ionic potential $V_{\text{ion}}(r) = -Z/r + V_{\text{loc}}^{(\text{static})}(r)$ behaves as $-1/r$ for $r \rightarrow \infty$, confirming that the core

screens $Z = 3$ to $Z_{\text{val}} = 1$. The key integrals are:

$$\int d^3\mathbf{r}' \frac{e^{-\alpha(r+r')}}{|\mathbf{r} - \mathbf{r}'|} = \frac{4\pi e^{-\alpha r}}{\alpha^2} u(r/2), \quad (\text{F4})$$

$$\int d^3\mathbf{r}' e^{-\alpha r'} u(r') = \frac{4\pi}{\alpha^2} \left[\frac{1}{18} u(3r/2) + \frac{\alpha}{27} \right]. \quad (\text{F5})$$

Nonlocal l -channel decomposition

The 1s exchange kernel has different strengths in each l channel. For $l = 0$, both exchange and projection contribute:

$$K^{l=0}(r, r') = -\frac{u_{1s}(r) u_{1s}(r')}{\max(r, r')} - u_{1s}(r) u_{1s}(r') [u(r) + u(r') - J], \quad (\text{F6})$$

where $u_{1s}(r) = 2\alpha^{3/2} r e^{-\alpha r}$.

For $l = 1$, only exchange contributes (the projection term vanishes by angular selection rules):

$$K^{l=1}(r, r') = -\frac{u_{1s}(r) u_{1s}(r')}{3} \frac{r_{<}}{r_{>}^2}. \quad (\text{F7})$$

The angular momentum decomposition of exchange follows from expanding $1/|\mathbf{r} - \mathbf{r}'| = \sum_l (4\pi/(2l+1)) r_{<}^l / r_{>}^{l+1} \sum_m Y_{lm}^*(\hat{r}) Y_{lm}(\hat{r}')$ and using orthogonality with the s -wave core.

We solve the radial Schrödinger equation on a uniform grid ($dr = 0.01$ Bohr, $r_{\text{max}} = 30$ Bohr) by discretizing the full nonlocal Hamiltonian as a dense matrix and diagonalizing. The eigenvalues are: $l = 0$ 1s: -4.499 Ha, $l = 0$ 2s: -0.199 Ha, $l = 1$ 2p: -0.127 Ha. The 2s eigenvalue is close to the all-electron Li value (-0.196 Ha), validating the EFT potential.

Kleinman–Bylander pseudopotential

We apply Troullier–Martins pseudization [17] to both l -channels with $r_c = 3.0$ Bohr: $u_{\text{ps}}^l(r) = r^{l+1} e^{p(r)}$ for $r < r_c$, where $p(r) = c_0 + c_2 r^2 + \dots + c_{12} r^{12}$ has 7 coefficients. Schrödinger inversion gives semilocal potentials:

$$V_{\text{sl}}^l(r) = \epsilon_l + \frac{u_{\text{ps}}''(r)}{2 u_{\text{ps}}(r)} - \frac{l(l+1)}{2r^2}. \quad (\text{F8})$$

The Kleinman–Bylander [26] form uses $V_{\text{local}} = V_{\text{sl}}^{l=1}$ and a single $l = 0$ projector $p_0(r) = \Delta V_0(r) \cdot R_{\text{ps}}^{l=0}(r)$ with $\Delta V_0 = V_{\text{sl}}^{l=0} - V_{\text{sl}}^{l=1}$ and $h_{00} = [\int R_{\text{ps}} \Delta V_0 R_{\text{ps}} r^2 dr]^{-1}$. The resulting pseudopotential has $Z_{\text{val}} = 1$, $l_{\text{max}} = 0$, and is implemented in DFTK [27].

Analytic frequency correction for lithium

The form factor [Eq. (7)] evaluates analytically:

$$f_K^{1s} = 8\sqrt{\pi/\alpha} A_K, \quad A_K = \frac{Q^2 + 33}{(Q^2 + 1)(Q^2 + 9)^2} - \frac{J}{\alpha} \frac{1}{(Q^2 + 1)^2}, \quad (\text{F9})$$

where $Q = K/\alpha$. For Li's single 1s channel, the core renormalization [Eq. (3)] reduces to

$$z_{\nu\mathbf{k}}^{\text{core}} = \frac{1}{1 + \left| \sum_{\mathbf{G}} c_{\nu\mathbf{k}}(\mathbf{G}) \frac{f_{|\mathbf{k}+\mathbf{G}|}^{1s}}{\Delta E_{1s}} \right|^2}, \quad (\text{F10})$$

with $\Delta E_{1s} = E_1 - E_0 = 2.770$ Ha. At Γ (valence band): $z^{\text{core}} \approx 0.75$.

Appendix G: Two-electron integrals in momentum space

The integral $\langle 1s \, 1s || 1s, \mathbf{K} \rangle$ evaluates to

$$\langle 1s \, 1s || 1s, \mathbf{K} \rangle = \frac{8\sqrt{\pi}}{\alpha^{1/2}} \frac{Q^2 + 33}{(Q^2 + 1)(Q^2 + 9)^2}, \quad Q = K/\alpha, \quad (\text{G1})$$

obtained by reducing to a radial integral via the convolution theorem and evaluating by contour integration.

The form factor relates to the ionization integral:

$$f_K^{1s} = \frac{1}{\sqrt{4\pi}} \left[\langle 1s \, 1s || 1s, \mathbf{K} \rangle - J \cdot \tilde{\psi}_{1s}(K) \right], \quad (\text{G2})$$

where $\tilde{\psi}_{1s}(K) = 8\sqrt{\pi} \alpha^{5/2} / (K^2 + \alpha^2)^2$.

The exchange integral reduces to a Feynman-parameter integral:

$$\langle 1s \, \mathbf{K} || 1s \, \mathbf{K}' \rangle = \frac{60\pi\alpha^5}{\pi^2} \int_0^1 dy \int_0^{1-y} dz \frac{yz}{\Delta(y, z)^{7/2}}, \quad (\text{G3})$$

where $\Delta = y(1-y)K^2 + z(1-z)K'^2 + 2yz \mathbf{K} \mathbf{K}' + (y+z)\alpha^2$.

Appendix H: Numerical pipeline for multi-electron cores

Step 1: Core orbitals. For Li, the 1s orbital is hydrogenic (Appendix F). For all other elements (Na, K, Ca, Mg, Al, Si), core orbitals are obtained from a self-consistent radial Kohn–Sham solver with Dirac exchange (no correlation) on a uniform grid ($\Delta r = 0.002$ Bohr, $r_{\text{max}} = 40$ Bohr). The solver is restricted to closed-shell occupations; the open-shell core-hole states are treated at the same level without spin polarization. This is an approximation: a spin-polarized or open-shell solver would give somewhat different ΔE_c and core orbitals, particularly for the outermost core shell. Since the correction

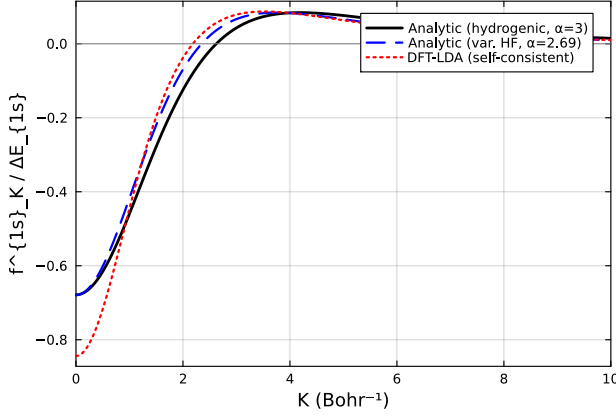


FIG. 4. $f_K^{1s}/\Delta E_{1s}$ for Li. Black solid: analytic with hydrogenic orbital ($\alpha = Z = 3$). Blue dashed: variational HF ($\alpha = Z - 5/16 = 2.6875$). Red dotted: self-consistent DFT-LDA core orbitals. All cross zero near $K \approx 2.5 \text{ Bohr}^{-1}$.

enters as a ratio $(f_0^c)^2/\Delta E_c^2$, partial error cancellation between numerator and denominator is expected, but the resulting uncertainty has not been quantified.

Step 2: Core excitation energies. ΔE_c is computed as a total energy difference: solving the atomic KS equations for the N_c -electron ground state and the (N_c-1) -electron state with a hole in orbital c , taking $\Delta E_c = E_{\text{hole}} - E_{\text{core}}$.

Step 3: Form factors. For each core s -orbital c :

$$f_K^c = \frac{\sqrt{4\pi}}{K} \sum_i u_c(r_i) [V_{H,c}(r_i) - J_c] \sin(Kr_i) \Delta r. \quad (\text{H1})$$

Core p -orbitals are omitted: $f_p(K) \propto j_1(Kr) \rightarrow 0$ near Γ .

Step 4: Coherent projection. f_K^c is precomputed on a fine grid ($K \in [0, 20] \text{ Bohr}^{-1}$, 2000 points) and interpolated. For each Bloch state $|\nu\mathbf{k}\rangle$, the core renormalization factor is computed via Eq. (3): $F_{\nu\mathbf{k}}^c = \sum_{\mathbf{G}} c_{\nu\mathbf{k}}(\mathbf{G}) f_{|\mathbf{k}+\mathbf{G}|}^c$, then $z_{\nu\mathbf{k}}^{\text{core}} = 1/(1 + \sum_c |F_{\nu\mathbf{k}}^c|^2/\Delta E_c^2)$.

Step 5: Quasiparticle energies via Eq. (2). The full post-SCF correction takes $\sim 0.2 \text{ s}$.

Validation for lithium

Figure 4 compares $f_K^{1s}/\Delta E_{1s}$ for three orbital models: the bare hydrogenic ($\alpha = Z$), the variational HF ($\alpha = Z - 5/16$), and self-consistent DFT-LDA. The hydrogenic and variational curves are nearly indistinguishable; the DFT-LDA curve is $\sim 25\%$ larger at $K = 0$ due to self-interaction error in the two-electron Li^+ core.

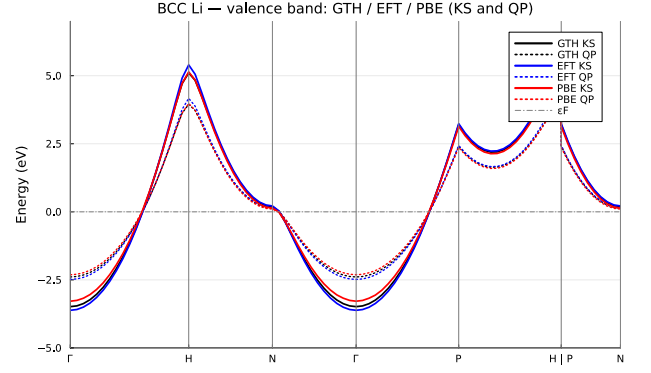


FIG. 5. Valence band structure of BCC Li from three PSP approaches: conventional GTH LDA (black), ab initio EFT nonlocal (blue), and PBE $Z_{\text{val}} = 3$ (red). Solid lines: KS eigenvalues; dashed lines: QP eigenvalues. The EFT QP narrows the Γ -point depth by 25%, from 3.48 to 2.62 eV, in good agreement with eDMFT (2.60 eV) [9].

PSP universality

All conventional $Z_{\text{val}} = 1$ pseudopotentials (GTH-LDA, HGH-LDA, HGH-PBE) give $\Gamma = -3.483 \text{ eV}$ for Li with spread $< 1 \text{ meV}$. For Na: $-3.267 \pm 0.001 \text{ eV}$. This universality confirms that the static potential is essentially unique for given Z_{val} .

PBE pseudopotential robustness

To test sensitivity to the static Kohn-Sham calculation, we repeat the EFT quasiparticle correction using PBE exchange-correlation together with PBE pseudopotentials from the CP2K GTH library, in place of the LDA functional and PSPs used throughout the main text. This simultaneously changes both the XC functional and the pseudopotential. For Mg, Al, and Si, PBE large-core PSPs have the same Z_{val} as the LDA PSPs. For Li, the PBE large-core PSP treats all three electrons explicitly ($Z_{\text{val}} = 3$). For Na, K, and Ca, the PBE library provides only small-core PSPs with larger Z_{val} (9, 9, and 10 respectively); these promote semicore s and p shells into the valence. When semicore bands are present, we apply the EFT correction only to the conduction band to avoid double counting with the now-explicit semicore states; semicore eigenvalues are left at their KS values.

Table II summarises the results. The bandwidth narrowing is remarkably stable: identical to within 0.2 percentage points for the same- Z_{val} cases (Mg, Al, Si), and within 1–2 percentage points even when Z_{val} changes.

Figure 6 shows the full band structures for all six elements (Li is shown separately in Fig. 5), overlaying LDA (solid) and PBE (dashed) results with and without the EFT quasiparticle correction. The near-perfect overlap confirms that the correction is insensitive to the static

TABLE II. EFT bandwidth narrowing (%) with LDA vs PBE exchange-correlation functionals and pseudopotentials. Z_v^{LDA} and Z_v^{PBE} denote the valence charges. For different- Z_v cases (Li, Na, K, Ca), only the conduction band is corrected.

	Z_v^{LDA}	Z_v^{PBE}	LDA (%)	PBE (%)
Li	1	3	10.1	10.4
Na	1	9	19.9	21.0
K	1	9	34.4	36.0
Ca	2	10	17.3	18.6
Mg	2	2	8.8	8.6
Al	3	3	4.3	4.1
Si	4	4	3.7	3.7

TABLE III. Na channel contributions. $f_0^c/\Delta E_c$ is the form-factor weight entering the coherent sum [Eq. (3)].

	ΔE_c (Ha)	f_0^c	$f_0^c/\Delta E_c$
1s	39.91	-1.05	-0.026
2s	2.70	+1.45	+0.537

Kohn–Sham setup.

Appendix I: Channel decomposition

Tables III and IV give the per-channel form-factor weights $f_0^c/\Delta E_c$ for Na and K. In both cases the outermost s -channel dominates; deeper channels are suppressed by their large ΔE_c . Table V summarises z_Γ^{core} for the valence band across all seven elements. The correction increases monotonically with decreasing ΔE_c , confirming the $1 - z^{\text{core}} \propto 1/\Delta E_c^2$ scaling.

Appendix J: Computational details and full band structures

All plane-wave DFT calculations use DFTK [27] with LDA exchange-correlation and Fermi–Dirac smearing at $T = 0.001$ Ha.

Lithium. BCC, $a = 6.632$ Bohr, $E_{\text{cut}} = 20$ Ha, $8 \times 8 \times 8$ k -grid. EFT nonlocal PSP: $r_c = 3.0$ Bohr for $l = 0, 1$; $\epsilon_{2s} = -0.199$ Ha, $\epsilon_{2p} = -0.127$ Ha; KB coupling $h_{00} \approx 24$ Ha (compensated by a small projector norm). Band structures with $n_{\text{bands}} = 4$, k -line density 20 Bohr.

Sodium. BCC, $a = 8.107$ Bohr, $E_{\text{cut}} = 15$ Ha. GTH LDA $Z_{\text{val}} = 1$ PSP. Atomic solver: $\Delta r = 0.002$ Bohr, $r_{\text{max}} = 40$ Bohr. Core excitation energies: $\Delta E_{1s} = 39.91$ Ha, $\Delta E_{2s} = 2.70$ Ha.

Potassium. BCC, $a = 9.874$ Bohr, same parameters as Na. Core excitation energy: $\Delta E_{3s} = 1.72$ Ha. f_K^c pre-

computed on 2000-point grid $[0, 20]$ Bohr $^{-1}$ with linear interpolation.

Calcium. FCC, $a = 10.545$ Bohr, $E_{\text{cut}} = 20$ Ha, $8 \times 8 \times 8$ k -grid. HGH LDA $Z_{\text{val}} = 2$ (large-core) PSP.

TABLE IV. K channel contributions. Same format as Table III.

	ΔE_c (Ha)	$f_0^c/\Delta E_c$
1s	~ 200	~ 0
2s	~ 12	~ 0
3s	1.72	-0.435

TABLE V. Core renormalization factor z_Γ^{core} across all elements studied [Eq. (3)].

	z_Γ^{core}	ΔE_c (Ha)	Dom. channel
Li	0.75	2.77	1s
Na	0.80	2.70	2s
K	0.66	1.72	3s
Ca	0.83	2.51	3s
Mg	0.91	4.07	2s
Al	0.96	5.69	2s
Si	0.98	7.57	2s

Same [Ar] core as K. Core excitation energy: $\Delta E_{3s} = 2.51$ Ha. The KS Γ -point depth narrows from 3.70 to 3.06 eV (17%).

Magnesium. HCP, $a = 6.066$ Bohr, $c/a = 1.624$, two-atom basis, $E_{\text{cut}} = 20$ Ha, $8 \times 8 \times 6$ k -grid. GTH LDA $Z_{\text{val}} = 2$ (large-core) PSP. Core excitation energy: $\Delta E_{2s} = 4.07$ Ha. Same [Ne] core as Na.

Aluminum. FCC, $a = 7.653$ Bohr, $E_{\text{cut}} = 20$ Ha, $8 \times 8 \times 8$ k -grid. GTH LDA $Z_{\text{val}} = 3$ (large-core) PSP. Core excitation energy: $\Delta E_{2s} = 5.69$ Ha. The KS Γ -point depth narrows from 11.18 to 10.70 eV (4%).

Silicon. Diamond cubic (FCC + two-atom basis), $a = 10.263$ Bohr, $E_{\text{cut}} = 20$ Ha, $6 \times 6 \times 6$ k -grid. GTH LDA $Z_{\text{val}} = 4$ (large-core) PSP. Core excitation energy: $\Delta E_{2s} = 7.57$ Ha. The KS Γ -point depth narrows from 12.26 to 11.81 eV (4%). The indirect band gap changes by only -0.002 eV; the EFT narrows all bands toward ϵ_F , reducing rather than opening the gap, as expected for a correction that captures core–valence screening but not the valence–valence self-energy responsible for gap opening (which requires *GW* or similar methods).

Figure 6 shows the full band structures for six elements with both LDA and PBE functionals (Li appears separately in Fig. 5), superseding individual per-element plots.

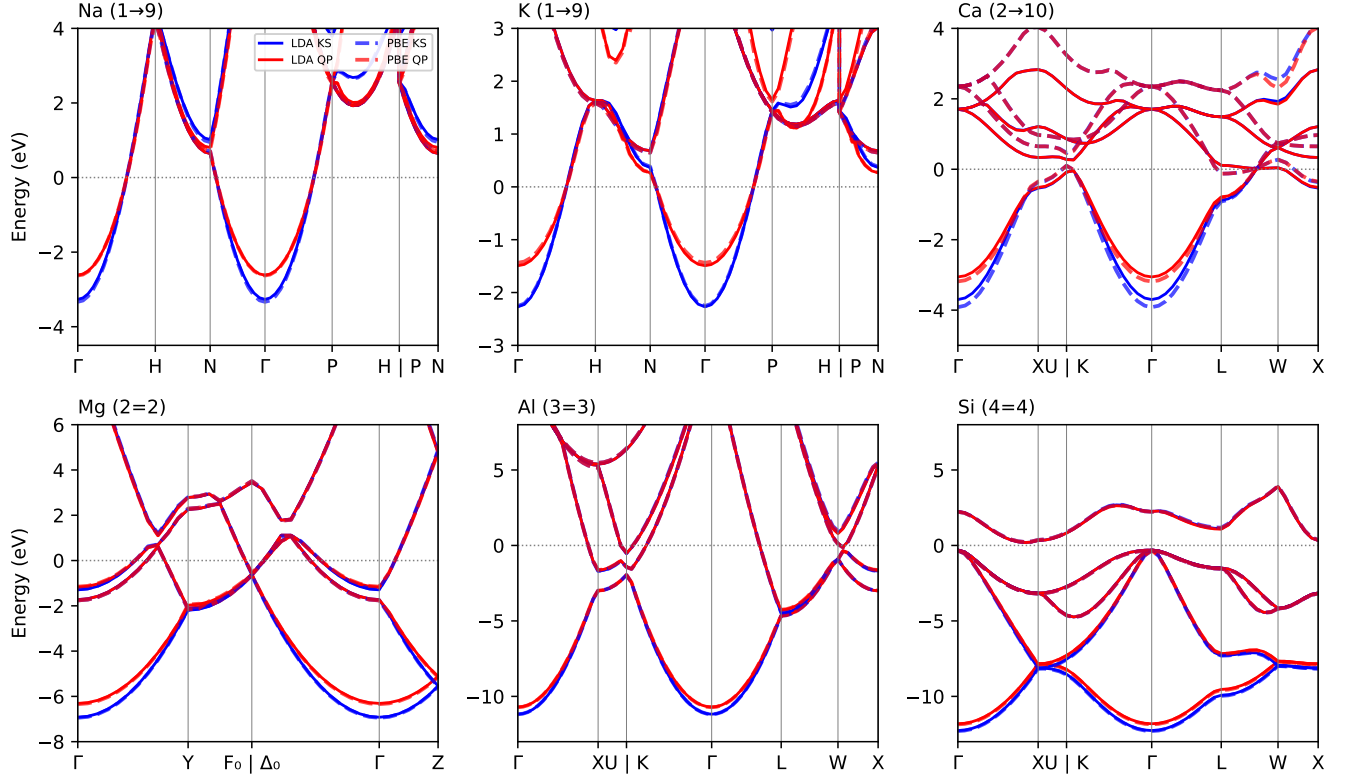


FIG. 6. LDA vs PBE band structures with EFT quasiparticle correction for six elements. Solid lines: LDA functional + PSP; dashed lines: PBE functional + PSP. Blue: KS eigenvalues; red: QP eigenvalues. Top row: different Z_{val} (Na 1→9, K 1→9, Ca 2→10); bottom row: same Z_{val} (Mg, Al, Si). For Na, K, and Ca, only the PBE conduction band is shown (semicore bands omitted for clarity). Li appears in Fig. 5.